# Ahmad Hossein Yazdani

**CONTACT INFORMATION**

Department of Computer Science
Faculty of Engineering
Virginia Polytechnic Institute and State University
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**AREAS OF INTEREST**

Design and Analysis of Algorithms, Artificial Intelligence, Machine Learning, Deep Learning, Storage Systems, File Systems, Kernel Programming, Enterprise Application Design, Computer Networks, Neuroscience, Adversarial Machine Learning, Natural Language Processing (NLP), Network Security, Back-end Development, Distributed computing, High performance computing, Cloud computing, Distributed Systems, High Performance Computing

**EDUCATION**

1. Virginia Polytechnic Institute and State University
   *PhD. In Computer Science*, August 2020-present

2. University of Tehran, Tehran, Iran
   *Bachelor of Science in Computer Software Engineering*, September 2015-July 2020

3. *Diploma at Allameh Helli High School,* June 2014

**RESEARCH EXPERIENCE**

**Research Assistant at Distributed System and Storage Lab at Virginia Tech. university**
*Supervised by Dr. Ali Butt*
*More information on https://dssl.cs.vt.edu/index.html*
*August 2020-present*

**RESEARCH PROJECTS**

1. *Stock price prediction using deep learning, finished in July 2020*
   *Under supervision of Professor Nasser Yazdani (As my undergraduate final project).*
   The code is available through GitHub

2. *Metis project as a research project that I have contribution to. It aims to do I/O caching for deep learning frameworks. It's still in progress.*

**NOTABLE PROJECTS**

*1. TOORLA Programming Language Compiler Development over JVM*

*2. ATALK Actor-based Programming Language Compiler Development over MIPS*

*3. Design and Implementation of House Rental Enterprise Application as Internet Engineering University Course Project by ReactJS and both Java Enterprise Application and NodeJS*

*4.. TCP Protocol Development under UDP as Computer Network University Course Project*

*5. Adding extra Synchronization mechanism to Linux kernel as Operating System University Course Project*

*6. MIPS Architecture Implementation on Cyclone II FPGA*

*7. Meeting Scheduler Enterprise Application as Software Engineering University Course Project*

*8. Design and Implementation of online Shopping Enterprise Application with Waterfall Software Development Methodology (Responsible for the back-end part by means of NodeJS)*

*9. Image classification using Spiking Neural Networks as my final Advanced Machine Learning course at Virginia Tech.*

*10. Development of a centralized cache on top of docker*

**NOTABLE PASSED COURSES**

- **Undergraduate Courses**

*1. Engineering Mathematics*
*2. Artificial Intelligence*
*3. Engineering Probability and Statistics*
*4. Computer Networks*
*5. Computer Workshop*
*6. Object Oriented Systems Design*
*7. Human Computer Interaction*
*8. Compiler Design and Implementation*
*9. Database Design*
*10. Computer Aided Digital System Design*
*11. Internet Engineering*
*12. Machines and Language Theory*
*13. Software Engineering*
*14. Parallel Programming*
*15. Computer Architecture Laboratory*
*16. Introduction to Software Testing*
*17. Formal Methods in Software Engineering*
*18. Object Oriented Systems Design*
*19. Human Computer Interaction*
*20. Computer Architecture*
*21. Microprocessors*
*22. Systems Analysis and Design*
*23. Calculus 1*
*24. Calculus 2*
*25. B.SC Project*

- **Graduate Courses**

1. Advanced Machine Learning
2. Graduate Operating System

**TEACHING ASSISTANTSHIP**

***Teaching Assistantship of Programming Languages and Compilers***
*University of Tehran*
*Fall 2018 - winter 2019*
*Presented by Dr. Fatemeh Ghassemi & Dr. Hossein Hojjat*
*As Homework Organizer and Course Project Grading TA*

***Teaching Assistantship of Programming Languages and Compilers***
*University of Tehran*
*Winter 2018 - spring 2019,*
*Presented by Dr. Hossein Hojjat*
*As Course Project Organizer*

***Teaching Assistantship of Formal Methods in Software Engineering***
*University of Tehran*
*Winter 2019 - spring 2020,*
*Presented by Dr. Fathieh Faghih*
*As both homework and project organizer*

***Teaching Assistantship of Artificial Intelligence***
*University of Tehran*
*Winter 2020 – summer 2020,*
*Presented by Dr. Hakimeh Fadaei*
*As both homework and project organizer*
*(Organized the project associated with Neural Networks and Deep Learning)*

***Teaching Assistantship of CS1114 (Introduction to Software Design)***
*Virginia Tech.*
*Fall 2020 – winter 2021,*
*Presented by Dr. Dwight Barnette*
*As Lab TA and Programming Assignment TA*

***Teaching Assistantship of CS3704 (Intermediate Software Design and Engineering)***
*Virginia Tech.*
*Winter 2021– Spring 2021,*
*Presented by Khadijah Al Safwan*
*As assignments' TA*

**ATTENDED CONTESTS**

***Attending Sharif University of Technology*** *contest hold in summer 2017*
***Attending codeforces contest*** *hold in 2017*

**INTERNSHIP EXPIERNECE**

***Internship at Router lab at University of Tehran*** *in summer 2017*
*Supervised by Dr. Nasser Yazdani*
*More information on https://digt.ir*

***Internship at Router lab at University of Tehran*** *in summer 2018*
*Supervised by Dr. Nasser Yazdani*

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| **LANGUAGE SKILLS** | Persian<br>English |
| **CERTIFICATES** | TOEFL, GRE |
| **FAVORITE PROGRAMMING LANGUAGES And FRAMEWORKS** | Programming Languages: C, C++, Java, Python, Go, HTML, CSS, JavaScript<br>HDL: Verilog, VHDL<br>Applications: Microsoft Office<br>Database: MySQL, SQLite, MS SQL<br>Machine Learning libraries and frameworks: sklearn, Pytorch, Tensorflow<br>Enterprise Application frameworks: Django, ReactJS, NodeJS<br>Containerization frameworks: Docker<br>Computer Network frameworks: GNS3<br>Document preparation: Latex |
| **HOBBIES** | Watching movies, Reading books, Quick Text Typing, Football, Chess |
| **REFERENCES** | Dr Ali Butt:<br>butta@vt.edu<br><br>Dr. Hoda Eldardiry:<br>hdardiry@vt.edu<br><br>Dr. Ahmad Khonsari:<br>aa_khonsari@ut.ac.ir<br><br>Dr. Hossein Hojjat:<br>hojjat@ut.ac.ir<br><br>Dr. Fatemeh Ghassemi:<br>fghassemi@ut.ac.ir<br><br>Dr. Faghih Fathieh:<br>f.faghih@ut.ac.ir<br><br>Dr. Nasser Yazdani:<br>yazdani@ut.ac.ir |

**LINKS**

1. https://www.linkedin.com/in/ahmad-hossein-yazdani-6560b3138/ *(LinkedIn profile)*

2. *https://github.com/Ayazdani1997 (GitHub profile)*

3. *https://gitlab.com/A.Yazdani (GitLab profile)*